

CS 673: Advanced Computer Vision

Course Code	CS 673
Course Name	Advanced Computer Vision
L-T-P-C	3-0-2-4
Prerequisites	EE511 Computer Vision or CS671 Deep Learning or EE608 Digital Image Processing or CS669 Pattern Recognition, or similar
Intended for	B.tech (Yd and 4111 Year), CSE, DSE and EE Students
Approval	54th BoA (May 2024)

Course Contents

- **Introduction to Computer Vision:** Get a conceptual overview of image classification, object localization, object detection, and image segmentation. Also be able to describe multi-label classification and distinguish between semantic segmentation and instance segmentation. In the rest of this course, you will apply PyTorch to build object detection and image segmentation models. (6 Hours)
- **Vision Models Classification:** Convolutional Neural Networks: Architectures, Convolution I Pooling Layers, spatial arrangement, layer patterns, layer sizing patterns, AlexNet/ZFNet/VGGNet/EfficientNet case studies, computational considerations, Transfer learning for Representation, domain adaptation, domain Generalization, open set, open world, curriculum learning (6 Hours)
- **Vision Models Object Detection and Tracking:** Overview of some popular object detection models, such as regional-CNN and ResNet-50. Use of object detection models in PyTorch, Transfer Learning for object detection: download your own models and configure them for training and build your own models for object detection. Using transfer learning, you will train a model to detect and localize. Model compression and real-time inference on Edge-devices. (9 Hours)
- **Vision Models Segmentation:** Types of semantic segmentation, instance-based models, Disadvantages of search window methods, RCNN-family models, Unet, DeepLab, Transformer based segmentation. (6 Hours)
- **Graphs and XAI for Vision:** Graph Representation/formulation for visual tasks, Deep Learning Models for Graphs, GradCam, GradCam++, Graph Model Explanation, LIME, GraphLIME (6 Hours)
- **Vision & Language Models:** Visual to Text: Recent methods for Text embeddings, Scene/Video Captioning, OCR, VQA, Video Description using RNNs, LSTMs. Transformers. Text to Visual: Text to image/video generation using V AEs, GANs, Transformers. (9 Hours)

Text Books

1. Richard Szeliski, **Computer Vision: Algorithms and Applications**.
2. David Forsyth and Jean Ponce, **Computer Vision: A Modern Approach**, 2nd Edition.

References

1. Ian Goodfellow, **Deep Learning**.
2. Trevor Hastie, Robert Tibshirani, and Jerome Friedman, **Elements of Statistical Learning**.
3. Richard Hartley and Andrew Zisserman, **Multiple View Geometry in Computer Vision**, 2nd Edition.